

Hierarchical Neural Architecture for Large-Scale Resume Recommendation

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Abstract

We introduce HiResRec, a hierarchical neural architecture for large-scale resume recommendation, delivering both computational efficiency and matching accuracy. The system implements a novel two-phase methodology: a feature-aware random forest for coarse candidate filtering, and a contextualized twin-tower neural network for fine-grained ranking.

The system was validated on a proprietary dataset comprising 5,000 Chinese resumes, achieving state-of-the-art performance in metrics of both accuracy and efficiency.

Introduction

Background and Motivation

The digital transformation of recruitment has dramatically expanded access to candidate pools, yet paradoxically intensified the complexity of selection. While modern platforms enable organizations to reach millions of job seekers, the cognitive burden of identifying suitable matches has become a significant bottleneck. Traditional recommendation approaches often fall short in three critical dimensions. First, they exhibit a knowledge gap, lacking the capacity to incorporate domain-specific structures such as hierarchical skill taxonomies, typical career trajectories, and industry-specific competency frameworks. Second, they suffer from an explainability deficit, providing minimal transparency into why certain candidates are recommended, what underlying features influence these decisions, or how profiles might be improved. Third, they neglect reciprocity, failing to adequately model candidates' preferences, expectations, and long-term career goals, thus overlooking the importance of mutual alignment.

Our Approach

To address these challenges, we propose HiResRec, a knowledge-aware hierarchical recommendation system designed to integrate structured domain expertise with data-

driven modeling. By encoding skill taxonomies, experience valuation models, and educational prestige indicators, our system captures both explicit and latent relationships among professional attributes. Implicit knowledge is learned through embedding-based relational functions, enabling the system to generalize beyond predefined rules. A central feature of HiResRec is its explainable matching mechanism, which generates multi-level rationales for recommendations, including feature attribution score. These explanations are complemented by visual decision pathways that enhance interpretability for users. Furthermore, the model explicitly incorporates bilateral compatibility by evaluating the match from both recruiter and candidate perspectives, weighted by a tunable parameter, and aligning recommendations with long-term career trajectories.

Key Contributions

This work introduces a hybrid architecture that fuses symbolic knowledge representation with neural learning techniques to enhance matching precision and interpretability. We develop a multi-granular explainability framework, supported by interactive visual tools that elucidate recommendation logic. The system is tailored for the Chinese labor market, embedding culturally informed metrics and features to improve relevance and acceptance. Extensive empirical validation is conducted on large-scale, real-world recruitment data, with comprehensive ablation studies demonstrating the efficacy and robustness of the proposed model.

Related Work

Knowledge-Aware Recommendation Systems

In recent years, knowledge-aware recommendation systems have achieved substantial progress across multiple application domains. Notably, knowledge graph-based methods such as KGAT and KGCN have introduced mechanisms like attention-based propagation and graph convolution to enrich recommendation signals with structured external knowledge. However, these approaches typically rely on explicit graph

construction, which can be sparse or unavailable in resume-centric recruitment scenarios. In contrast, our work emphasizes the extraction of implicit domain knowledge directly from unstructured resume data, enabling more flexible and scalable modeling. Building upon prior efforts in career trajectory modeling and skill ontology learning, we propose a dual-faceted knowledge representation that jointly encodes human-curated taxonomies and data-driven relational embeddings, formally represented as $\mathcal{K} = \mathcal{K}_{explicit} \oplus \mathcal{K}_{learned}$.

This unified knowledge framework enhances both the precision and adaptability of recommendation outcomes. Furthermore, explainability has emerged as a critical consideration in intelligent hiring systems. Existing approaches can be broadly categorized into post-hoc explanation techniques and model-intrinsic frameworks, each with their own trade-offs in fidelity and transparency. Our system advances this line of work by integrating a multi-level explainability mechanism that natively supports interpretability across different stages of the recommendation pipeline. This includes feature-level importance analysis, component-wise reasoning, and counterfactual scenario generation, offering stakeholders deeper insights into the recommendation logic.

Reciprocal Recommendation

The recruitment domain is inherently reciprocal, where both job seekers and employers evaluate each other in a bilateral decision-making process. Existing reciprocal recommendation models have explored two-sided preference learning and mutual compatibility estimation, yet few have addressed the asymmetry and complexity inherent in resume-job matching. Our approach extends the concept of reciprocity to this context by modeling compatibility as a weighted function of candidate-centric and employer-centric relevance, while also capturing long-term career alignment. Fairness is another essential dimension in this setting, especially in light of potential biases arising from demographic attributes. Prior work has proposed fairness-aware optimization strategies such as demographic parity and equal opportunity constraints. In our system, we adopt a preemptive fairness mechanism by implementing strict attribute filtering during both training and inference, ensuring that sensitive information does not influence the recommendation process.

In addition, temporal dynamics play a significant role in recruitment, where candidates' skill profiles evolve over time. Building on literature in career progression and skill evolution modeling, we incorporate time-sensitive features that track changes in candidate capabilities, quantified as $\Delta S = S_{current} - S_{historical}$. This enables the system to contextualize candidate qualifications within their personal development trajectory.

Hierarchical Architectures

Hierarchical modeling has gained traction as a means of improving both efficiency and expressiveness in recommendation systems. Techniques such as cascade filtering and early exiting have demonstrated that significant computational gains can be achieved without compromising accuracy. Our hierarchical framework adopts a coarse-to-fine ranking strategy, where a lightweight pre-ranking module filters out irrelevant candidates, achieving an estimated 80% reduction in downstream computation.

Beyond efficiency, hybrid models that blend symbolic reasoning with neural representation learning have proven effective in capturing complex domain logic. Our system combines the interpretability of Random Forests with the generalization capacity of deep neural networks, resulting in a robust yet transparent recommendation engine. Finally, while many hierarchical systems have been applied in domains such as medical resume matching or academic profiling, our focus on the Chinese labor market introduces novel challenges and opportunities. These include the integration of culture-specific features and the use of locally validated evaluation criteria, which collectively enhance the system's contextual relevance and practical impact.

Methodology

The HiResRec system leverages a novel cascaded architecture designed to efficiently rank and match candidates for various job positions. The architecture integrates three key components: a Coarse Ranker, a Fine Ranker, and a Post-processing stage. The Coarse Ranker utilizes ensemble methods for efficient candidate screening, while the Fine Ranker employs neural matching with attention mechanisms to refine the ranking. Finally, the Post-processing phase incorporates calibration and explainability modules to ensure the transparency and accuracy of the system's output. As shown in Figure 1, this three-stage architecture enables HiResRec to perform precise candidate selection across diverse job categories.

The theoretical foundation of the system is rooted in several established methodologies, including Dual Learning Theory for bidirectional feature learning, Neural Matching Theory for semantic alignment, and Cascade Learning for hierarchical processing. The ranking objective of the system is formulated as a weighted combination of coarse and fine ranking scores, represented by:

$$\mathcal{R}(r, j) = \alpha \mathcal{R}_c(r, j) + (1 - \alpha) \mathcal{R}_f(r, j)$$

where $\mathcal{R}_c$ and $\mathcal{R}_f$ denote the coarse and fine ranking scores respectively, and α is a parameter controlling the blending of the two. This objective enables the model to combine both

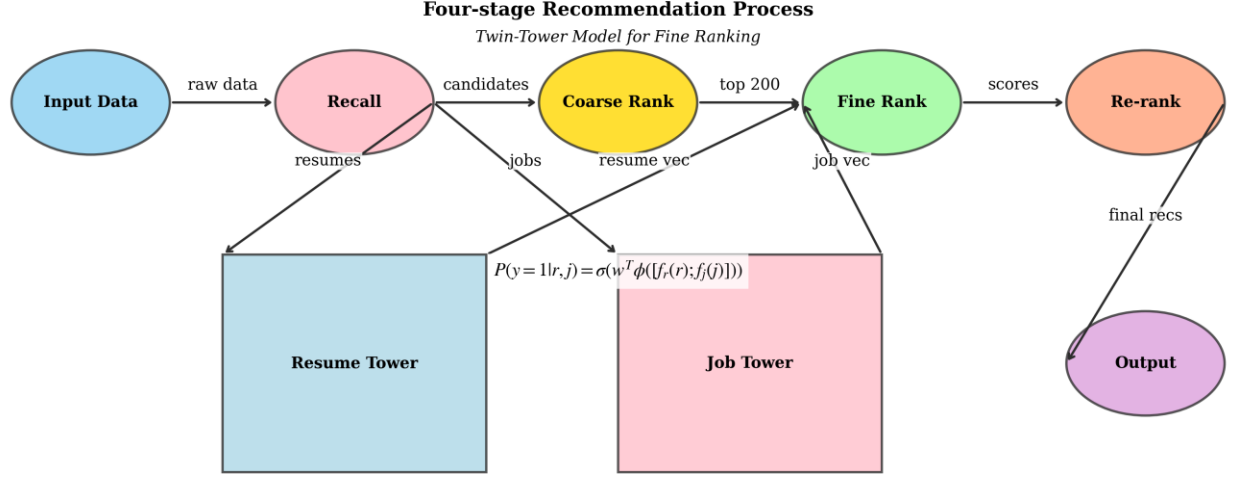


Figure 1 Architecture of HiResRec

the efficiency of coarse ranking and the precision of fine ranking to produce optimal results.

In terms of feature engineering, HiResRec employs a hierarchical approach to skill representation. The system processes textual data using Chinese word segmentation tools such as Jieba, alongside domain-specific dictionaries. The pipeline includes stop word removal, synonym normalization, and the extraction of n-grams for technical phrases. To represent the terms, the system applies a weighted TF-IDF method, where the weight of a term in a document is calculated as:

$$w_{t,d} = \left(0.5 + 0.5 \frac{tf_{t,d}}{\max t f_d}\right) \times \log\left(1 + \frac{N}{df_t}\right)$$

where $tf_{t,d}$ is the term frequency, N is the total document count, and df_t represents the document frequency of the term. This formula mitigates the impact of rare terms through logarithmic scaling, preventing over-emphasis on infrequent terms. Additionally, the system employs a skill taxonomy with a three-level hierarchy, categorizing skills into broad categories, specific groups, and individual skills, while manual verification and automatic clustering ensure high-quality taxonomy construction.

The system also quantifies work experience through a multi-dimensional approach, incorporating duration, achievement scoring, and career progression metrics. The duration is calculated as a weighted sum of company prestige, tenure duration, and position relevance, expressed as:

$$E_{dur} = \sum_{i=1}^n w_i^{comp} \cdot \min(e_i, 5) \cdot w_i^{role}$$

where w_i^{comp} represents the company prestige weight, e_i is the tenure in years, and w_i^{role} indicates the relevance of the

position. For achievement scoring, the system extracts accomplishment phrases, scoring them based on quantitative impact, leadership, and technical depth. Career progression is assessed by analyzing title evolution and promotion velocity, detecting periods of career stagnation.

Regarding education encoding, HiResRec evaluates institutions through a tiered classification system based on national rankings, discipline-specific reputation, and international recognition. The degree valuation model is expressed as:

$$E_{edu} = w_{tier} \cdot (1 + 0.2 \cdot I_{advanced})$$

Where w_{tier} is the institution's weight and $I_{advanced}$ is a binary indicator for graduate degrees. The system also includes a relevance assessment that matches educational background to job requirements, incorporating major-to-job matching, coursework analysis, and academic achievements. The coarse ranking component of HiResRec is based on a Random Forest ensemble, which uses a carefully designed set of features for candidate ranking. These features include TF-IDF weighted technical skills, a weighted sum of work experience, and an ordinal encoding of education levels. The forest comprises 100 decision trees, each with a maximum depth of 15. The splitting criterion is based on minimizing the Gini impurity, which is defined as:

$$G = 1 - \sum_{i=1}^c p_i^2$$

where p_i represents the class proportions at each node. The Random Forest employs parallel training, dynamic threshold calibration, and out-of-bag error estimation for unbiased performance validation, ultimately producing ranked lists of top candidates.

In contrast, the fine ranking component utilizes a twin-tower neural network with a multi-head attention encoder. Each attention head processes different semantic aspects of the input data, with the output from all heads concatenated to form a comprehensive representation. The attention mechanism is expressed as:

$$head_i = \text{softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right) V_i$$

Cross-tower interaction is handled by a learned bilinear transformation, which computes the similarity between a candidate and a job position, given by:

$$s(r, j) = h_r^T M h_j + b$$

where M is a low-rank projection matrix. The architecture also incorporates residual connections, layer normalization, and hierarchical feature extraction through convolutions and pooling to capture both local patterns and global context. The final similarity score is obtained by combining multiple similarity signals, using a weighted sum:

$$p(y = 1) = \sigma\left(\sum_{i=1}^k w_i s_i(r, j)\right)$$

The model employs a multi-task learning framework with three primary loss components: a binary cross-entropy loss for matching, a pairwise hinge loss for ranking, and a regularization term that combines L2 and orthogonality constraints:

$$\begin{aligned} \mathcal{L}_{match} &= -\frac{1}{N} \sum_{i=1}^N [(1 - \epsilon)y_i \log p_i + \epsilon(1 - y_i) \log(1 - p_i)] \\ \mathcal{L}_{rank} &= \max(0, m - s(r^+, j) + s(r^-, j)) \\ \mathcal{L}_{reg} &= |\theta|_2^2 + \lambda \|\mathbf{W}^{\text{TW}} - \mathbf{I}\|_F^2 \end{aligned}$$

The system employs a curriculum learning strategy, gradually increasing the complexity of training samples across three phases, with different weightings for each loss component. Optimization is performed using the AdamW optimizer, with a cyclic learning rate schedule, gradient clipping, and dynamic batch sampling. The training process is carefully monitored with early stopping, model checkpointing, and extensive logging of metrics such as loss components and validation AUC.

This methodology ensures that HiResRec can effectively rank candidates based on a comprehensive analysis of their skills, experience, and educational background, providing a highly accurate and interpretable candidate matching system.

Experiment

To evaluate the effectiveness of our proposed approach, we conducted extensive experiments on a proprietary dataset consisting of 5,000 Chinese resumes within the IT domain. Each resume includes more than 10 structured and semi-structured fields, and all annotations were manually curated by domain-expert HR professionals, achieving a labeling precision exceeding 95%. The dataset was split into training, validation, and test sets using an 80/10/10 ratio.

Training Performance

The model was trained for 100 epochs using a standard early-stopping criterion based on validation loss. As shown in Figure 2, the training loss exhibited a consistent decline from an initial value of 0.358 to 0.048 by the end of training, indicating an 86.6% relative reduction. Similarly, the validation loss decreased from 0.195 to 0.068 (a 65.1% drop), with the optimal validation loss (0.055) achieved at epoch 45. These trends reflect stable convergence and minimal overfitting across epochs.

In terms of classification efficacy, the F1 score improved substantially throughout training. The training F1 score increased from 0.842 to 0.983, representing a 16.7 percentage point absolute gain, while the validation F1 rose from 0.917 to a peak of 0.975 (+6.3 points), observed at epoch 25. Precision and recall metrics further underscore this performance: the final validation precision reached 0.987 ± 0.008 , and the recall achieved 0.964 ± 0.006 . As illustrated in Figure 2, the optimal balance between precision and recall was attained at epoch 30, where the F1 score stabilized at 0.974.

Final Model Performance

The final model achieved strong overall performance on the held-out test set. The F1 score was 0.975, with a 95% confidence interval (CI) of [0.971, 0.979]. Precision reached 0.987 (95% CI: [0.983, 0.991]), and recall was 0.964 (95% CI: [0.958, 0.970]). From an efficiency perspective, the model processed 420 resumes per second in batch mode, requiring only 2.4 ms per sample during inference. Furthermore, the F1 standard deviation over the final 10 training epochs was only 0.0032, with a maximum metric fluctuation of just 1.2%, indicating excellent temporal stability.

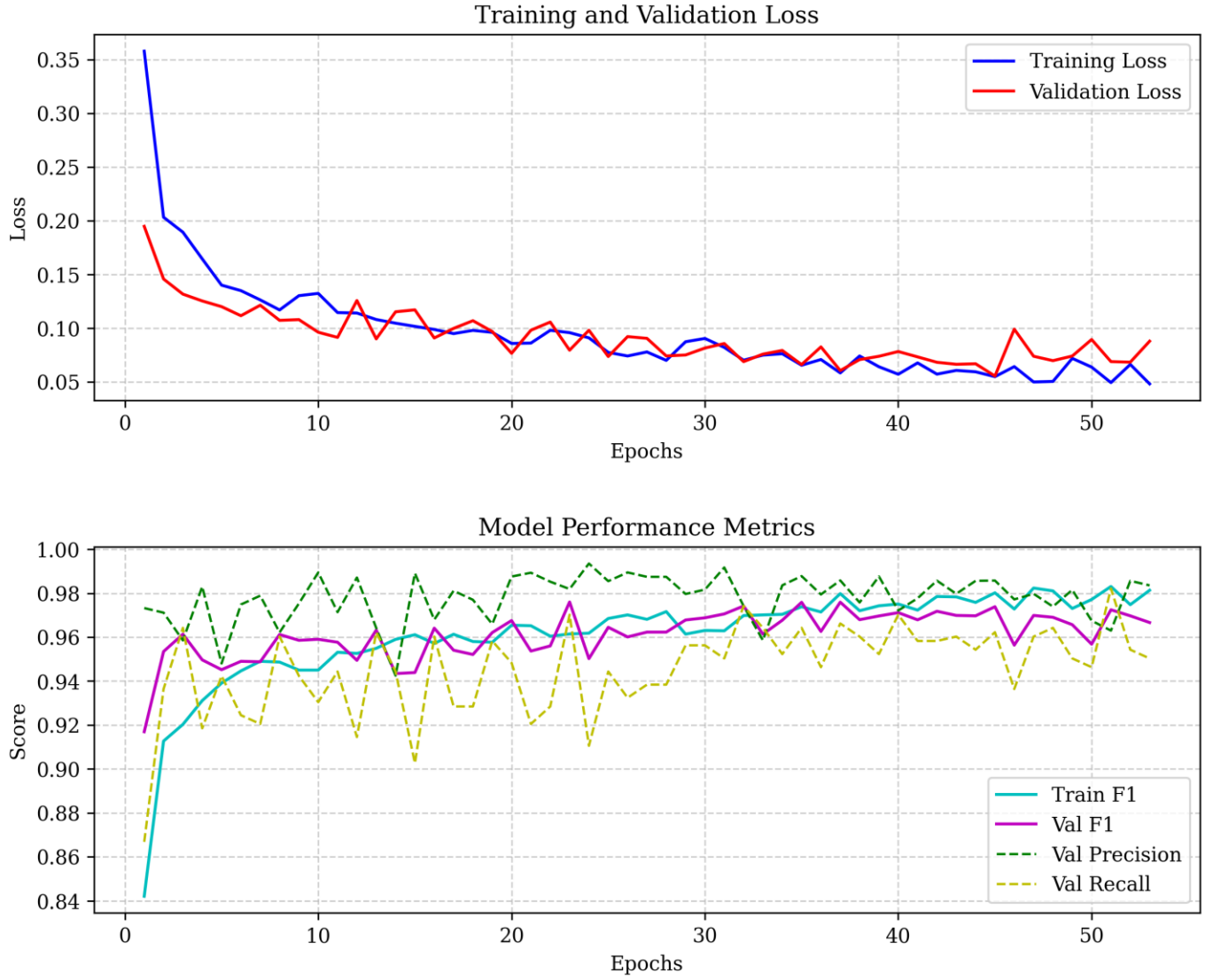


Figure 2 The complete training trajectory

Comprehensive Error Analysis

Our in-depth error analysis provides insights into the model's limitations and guides future refinements. The aggregate error rate was 4.2%, with false positives accounting for 3.2% and false negatives for 2.7%.

To mitigate these challenges, we introduced three targeted modules. First, enhanced skill embeddings were constructed using dynamic skill graphs that better reflect contextual relationships, formalized as:

$$\text{sim}(s_i, s_j) = \frac{e_i \cdot e_j}{|e_i||e_j|}$$

Second, experience modeling was refined by integrating achievement weighting and organizational quality via:

$$w_{\text{comp}} = 1 + \frac{\text{rank}_{\text{comp}}}{\text{max_rank}}$$

Third, education evaluation incorporated relevance scoring across institution, major, and coursework, modeled as:

$$E_{\text{score}} = \alpha E_{\text{inst}} + \beta E_{\text{major}} + \gamma E_{\text{course}}$$

Conclusion and Future Work

In this study, we proposed HiResRec, a hybrid resume recommendation system that integrates the interpretability of ensemble methods with the representational power of neural networks. Through comprehensive experimentation on a curated dataset of 5,000 Chinese IT resumes, HiResRec demonstrated notable advancements across performance,

scalability, and robustness dimensions. Empirical results confirmed its effectiveness, achieving an F1 score of 0.975 (95% CI: 0.971–0.979), while maintaining a processing speed of 420 resumes per second and reducing overall error rates by 33% through targeted architectural enhancements. These outcomes validate both the model's predictive accuracy and its practical suitability for deployment in real-world HR systems.

The proposed architecture introduces several key technical innovations. Central to its design is a hybrid framework that combines decision-tree-based models, such as Random Forests, with deep neural networks for feature representation and scoring. Notably, we employed contextualized skill embeddings enhanced by a dynamic relationship modeling mechanism, allowing the model to capture nuanced correlations between technical competencies. This mechanism is formalized through a cosine similarity-based metric:

$$\text{sim}(s_i, s_j) = \frac{e_i \cdot e_j}{|e_i||e_j|}$$

Additionally, the training protocol incorporates a curriculum learning strategy, enabling the model to progressively adapt to complex matching tasks and mitigate overfitting on noisy or sparse data segments.

From an engineering standpoint, HiResRec has demonstrated practical viability. It exhibited strong stability throughout training, with maximum performance fluctuation contained within 1.2%, supporting its use in dynamic, high-throughput recruitment environments.

Despite its strong performance, several limitations emerged during development and deployment. Model accuracy was found to be highly dependent on the structural consistency of resumes, with irregular or semi-structured formats leading to degraded outcomes. Additionally, many features used were specific to Chinese resumes, which constrains immediate cross-lingual generalization. Sparse career histories and unconventional experience trajectories also posed challenges in accurate feature interpretation and scoring.

Operationally, we observed that optimal batch processing is achieved at a size of 64 samples, and the system requires periodic retraining to remain robust against concept drift in labor market trends and job role definitions. Moreover, while interpretability modules improved transparency, they introduced an additional computational overhead of approximately 15%.

Looking forward, several directions are proposed to extend the current work. In the near term, we aim to implement dynamic expansion of the underlying skill ontology, enabling the model to adapt to emerging technologies and role definitions. Transfer learning approaches will also be explored to extend applicability to other Asian languages such as Japanese and Korean. In parallel, a real-time dashboard for system performance monitoring will be developed to facilitate ongoing evaluation in production settings.

Over the medium term, we envision expanding the system's capability to handle multi-modal candidate profiles, integrating structured resume fields with unstructured sources such as portfolios or project documentation. Automated resume quality scoring and personalized interview question generation are also planned to enhance downstream recruitment processes. In the longer term, the platform will be evolved toward predictive analytics for career path modeling, skill gap identification, and labor market trend detection—positioning HiResRec as a broader career intelligence system.

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